Atharv Mahajan

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Professional Summary

Machine Learning Engineer who builds first and learns by doing. Over the past year, shipped production ML systems across computer vision, NLP, and financial risk modeling, delivering end-to-end solutions that improved model performance and reliability in real-world settings. Comfortable across the full stack: model training, AWS deployment, and user-facing interfaces. Looking for a role where the problems are hard and shipping things that work actually matters.

Technical Skills

Core ML & Data: Python, NumPy, Pandas, scikit-learn, TensorFlow, PyTorch, XGBoost, NLTK
Backend & MLOps: FastAPI, PostgreSQL, MySQL, Git, GitHub, LangChain, n8n
AWS: EC2, S3, Lambda, SageMaker, Amazon Bedrock
Frontend (Secondary): JavaScript, React

Experience

Leads-Campaign | *Machine Learning Intern* — *March 2025 – March 2026*

- Engineered a Random Forest classifier to automate car insurance claim categorization, achieving **96.4% accuracy** on a dataset of **300+ monthly claims** and reducing manual review time by **15%.**
- Built an XGBoost credit scoring engine for financial risk assessment, improving F1-score from **0.84** (logistic regression baseline) **to 0.92** on a **4,000-applicant dataset**, and surfaced model interpretability via SHAP.
- Deployed an n8n automation pipeline for CRM data entry, eliminating **25% of input errors** and reducing processing time by **15%**, removing manual bottlenecks across the sales workflow.

Projects

lungcare.ai | *React, TensorFlow, Google ViT, Hugging Face* — lungcareai.vercel.app

- Developed a full-stack web application enabling healthcare professionals to detect and classify lung cancer from histopathological images, deployed on Vercel with Hugging Face Spaces for inference.
- Fine-tuned Google's Vision Transformer (ViT) on the LC25000 dataset, achieving **98% classification accuracy** across 5 lung and colon tissue categories.
- Optimized a TensorFlow `tf.data` preprocessing and augmentation pipeline for a **25,000-image corpus**, reducing model training time by **15%.**

AI-Fitness-Tracker | *Streamlit, OpenCV, MediaPipe, Python, Gemini API* — fitness-tracker-cv.streamlit.app

- Built a computer vision fitness tool using MediaPipe and OpenCV to track form and count reps across **15+ exercises** including squats, push-ups, and deadlifts.
- Engineered a real-time pose estimation pipeline achieving **30 FPS** on standard webcam hardware, with joint-angle calculations driving live form correction feedback.
- Integrated Gemini 2.5 Flash via API with structured prompt engineering to generate corrective workout guidance from real-time pose data, validated across **50 test sessions.**

Certifications

- **AWS Certified Machine Learning Engineer – Associate**
  Covers end-to-end ML workflows on AWS: SageMaker pipelines, model deployment, and large-scale feature engineering and data preparation. Applied in building and deploying production ML models.
- **AWS Certified Solutions Architect – Associate**
  Covers cloud infrastructure design across EC2, S3, Lambda, VPC, and IAM.
- **LangChain Academy – Introduction to LangGraph**
  Covers building stateful, multi-step LLM agents using LangGraph's graph-based orchestration framework, including tool use, memory, and conditional routing between agent nodes.

Education

Acropolis Institute of Technology & Research — Indore, M.P.
Bachelor of Technology in Computer Science and Engineering — *Expected May 2026*